

 OVERVIEW

OV80i



#4 Camera 56959
Tail

Inspection Problem

The part is a long, formed sheet-metal tail rail held in a fixture on the line. Three conditions have to be confirmed on every part: cracks anywhere along the body of the rail, presence of six punched holes along the flange, and the quality of the flared "horn" end (missing-end / surface condition). Inspection runs in-line, with the plant PLC triggering the camera on each cycle, so the verdict must be produced automatically and returned to the controls system.

Inspection System Configuration

A single Overview AI camera runs one recipe with three AI models: pixel-level crack segmentation across five zones tiling the rail, a hole-presence classifier on six hole locations, and a horn-quality (missing end) classifier on the end feature. Both classifiers reached 100% training accuracy — horn quality on 30 labeled examples (23 Pass / 7 Fail) and hole presence on 83 labeled examples (74 pass / 9 fail) — while the crack model trained down to 0.028 loss. The camera combines all three results into one pass/fail verdict and sends it to the plant PLC over industrial Ethernet on every cycle, with a detailed result report also forwarded to the cell's master controller.

#4 Camera 56959 Tail

A fixtured sheet-metal tail rail is inspected for cracks along its body, presence of six punched holes, and the condition of the flared horn end.



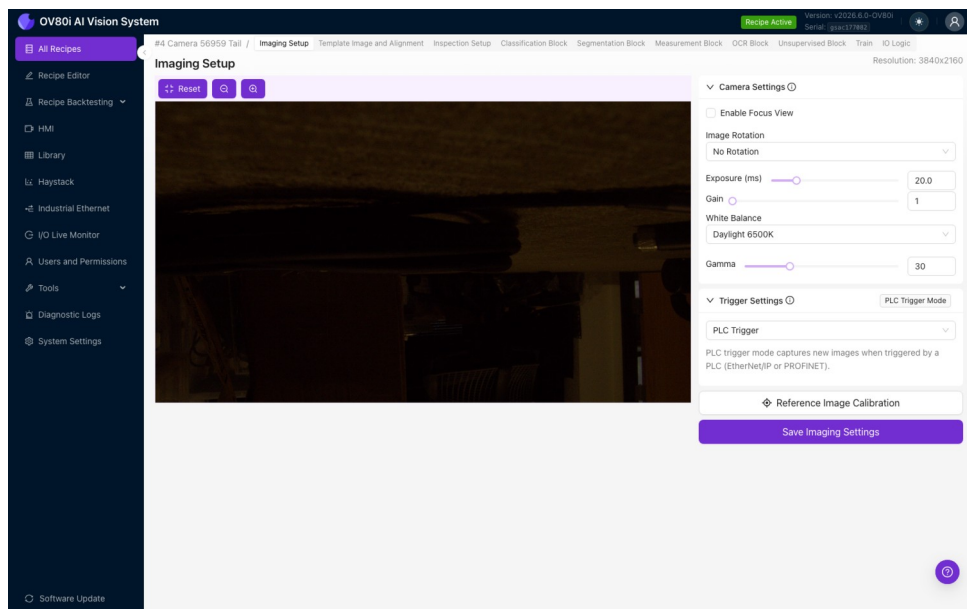


OVERVIEW

Configuring OV80i

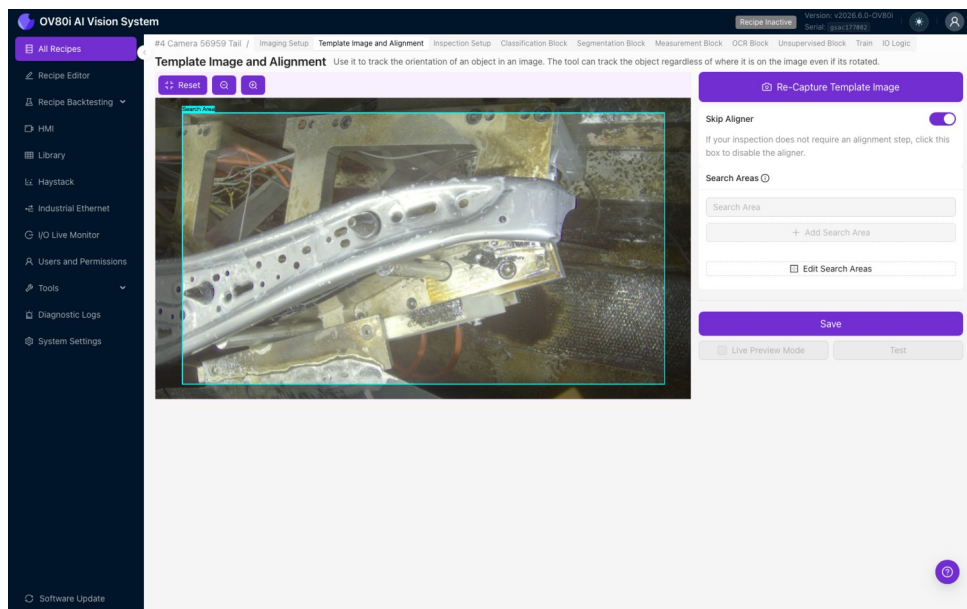
Power of a vision system, ease-of-use of a vision sensor.

Step 1: Setup Camera and Lighting



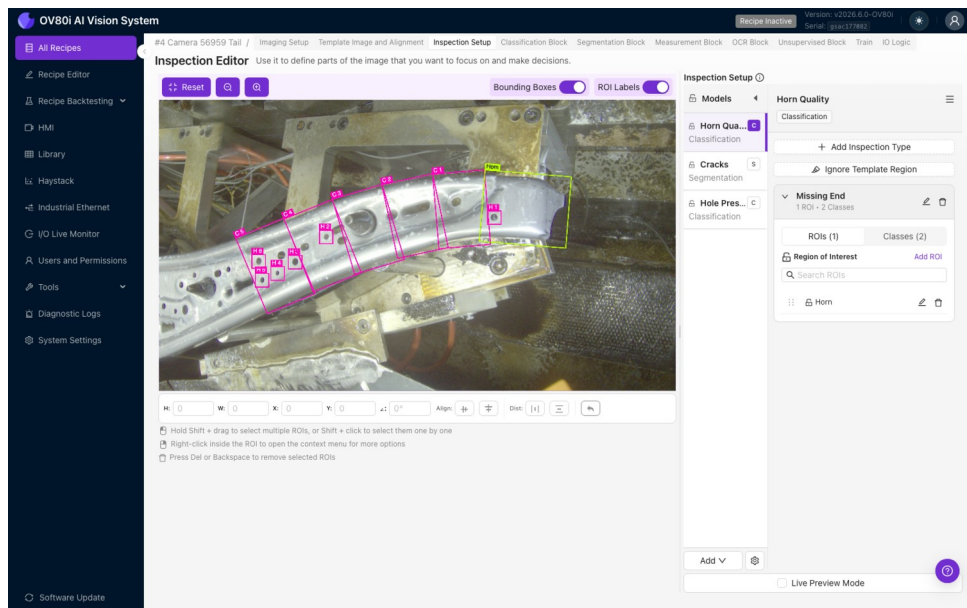
A single 4K (3840x2160) color camera images the full length of the rail in its fixture under even lighting. The plant PLC triggers a capture on every production cycle.

Step 2: Setup Template Image and Aligner



Because the rail is held in a hard fixture and repeats position, the alignment step is switched off and no search areas are used. Inspection regions are applied directly to each capture.

Step 3: Setup Inspection Regions



One region, "Horn", frames the flared end of the rail. It is the single area judged for the missing-end condition.

Step 4: Setup classifier inspection

OV80i AI Vision System

Recipe inactive | Version: v2028.6.5-OV80i | Serial: 3901171901

#4 Camera 56959 Tail / Imaging Setup Template Image and Alignment Inspection Setup **Classification Block** Segmentation Block Measurement Block OCR Block Unsupervised Block Train IO Logic

Classification
Use it to categorize an image into different classes based on its visual characteristics.

Reset Realign Part Bounding Boxes ROI Labels Delete

Skip Aligner is Enabled
Alignment is being skipped. This may result in inconsistent ROI positioning and reduced model accuracy. Consider disabling skip aligner for better training data quality.

Return to Live Generate Predictions

Navigation Source Capture: 39 of 39 Go

View All ROIs Previous Next

Capture ID: 113028 (Headless)
Capture Time: 2026-05-21 12:16:37:000 / 2 months ago
Capture Notes:
Cloned from recipe: Camera 4 David's Version - Capture ID: 113017

Inspection Types

Inspection Type	Pass	Fail
Hole Presence - Hole Presence	74	9
Missing End - Horn Quality	23	7

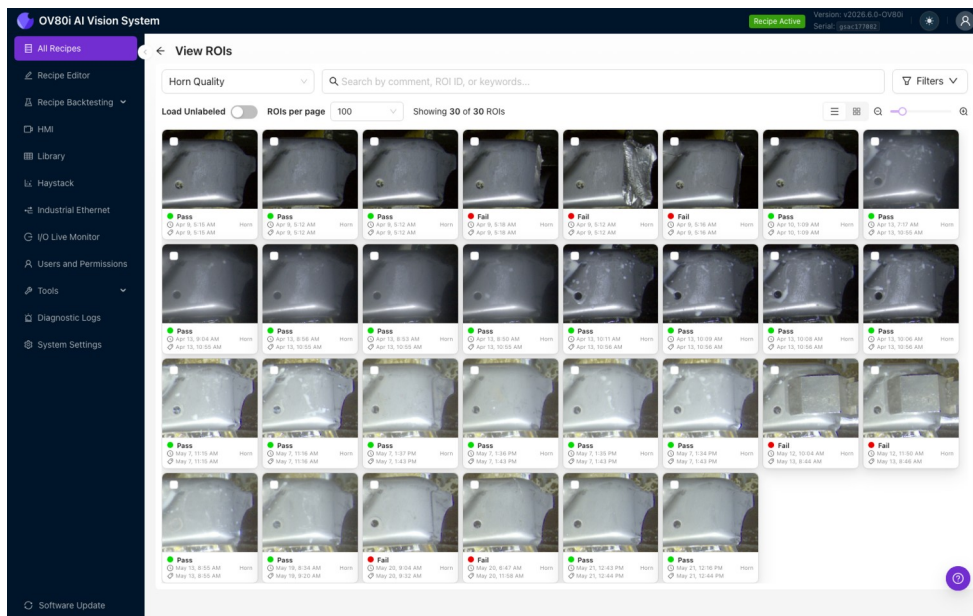
Train

Live Preview Mode Test

A two-class classifier that judges the horn end for a Missing End condition. Classes are Pass (pass_missing_end) and Fail (fail_missing_end).

All labeled regions — Horn Quality

The gallery shows all 30 labeled crops of the Horn region, each tagged with its class and capture time. 23 are labeled Pass and 7 Fail, with fails showing deformed, discolored or rough end surfaces. This is the reviewed dataset that defines what a good versus bad horn end looks like.



Training settings — Horn Quality

Training mode: Accurate

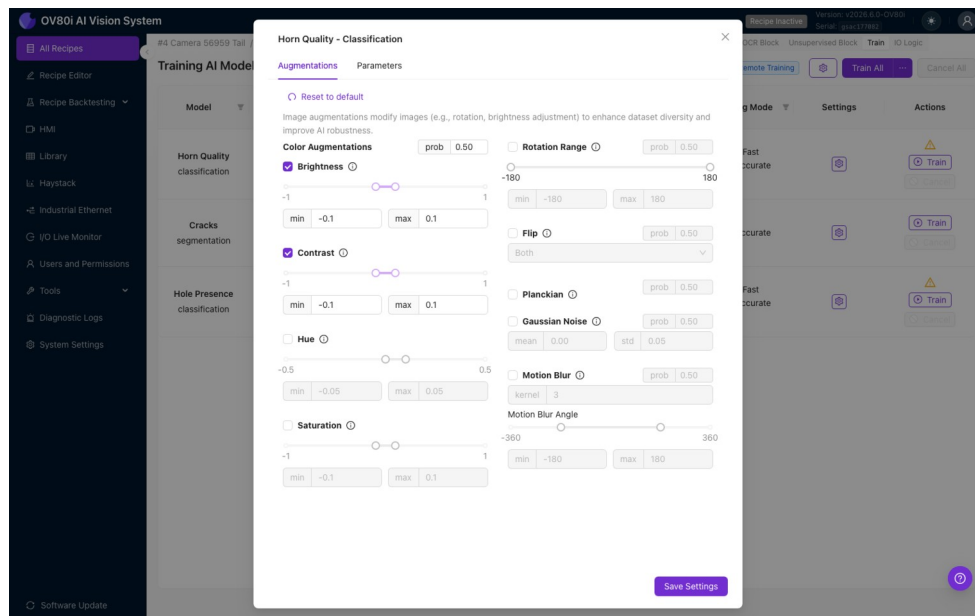
Color augmentation probability: 0.50

Brightness augmentation: enabled, -0.1 to 0.1

Contrast augmentation: enabled, -0.1 to 0.1

Hue and saturation: disabled

Rotation, flip, noise and motion blur: disabled



Training report — Horn Quality

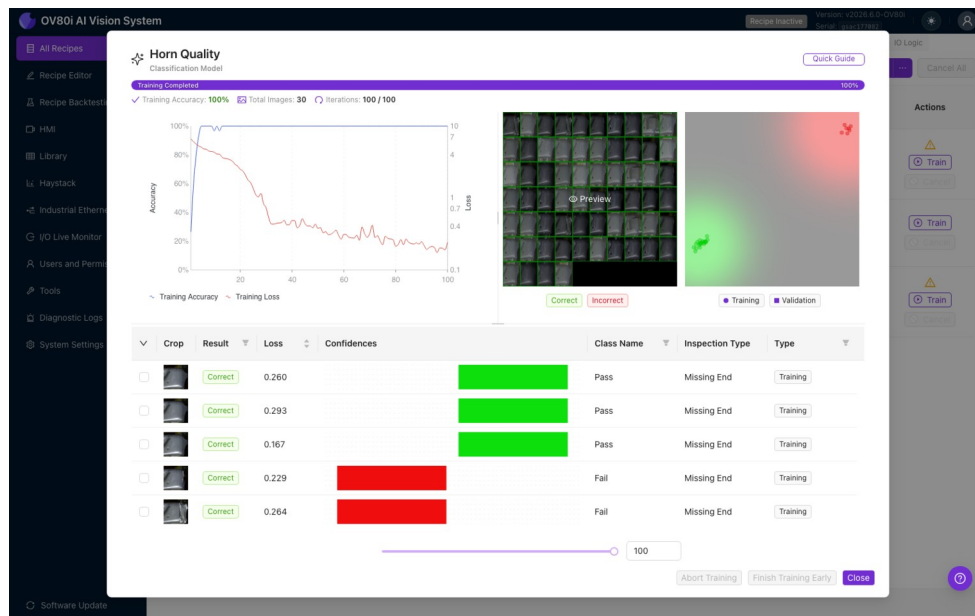
Training accuracy: 100%

Training loss: 0.240

Training images: 30

Iterations: 100 / 100

Labels: 23 Pass, 7 Fail



Results

100%

Training Accuracy

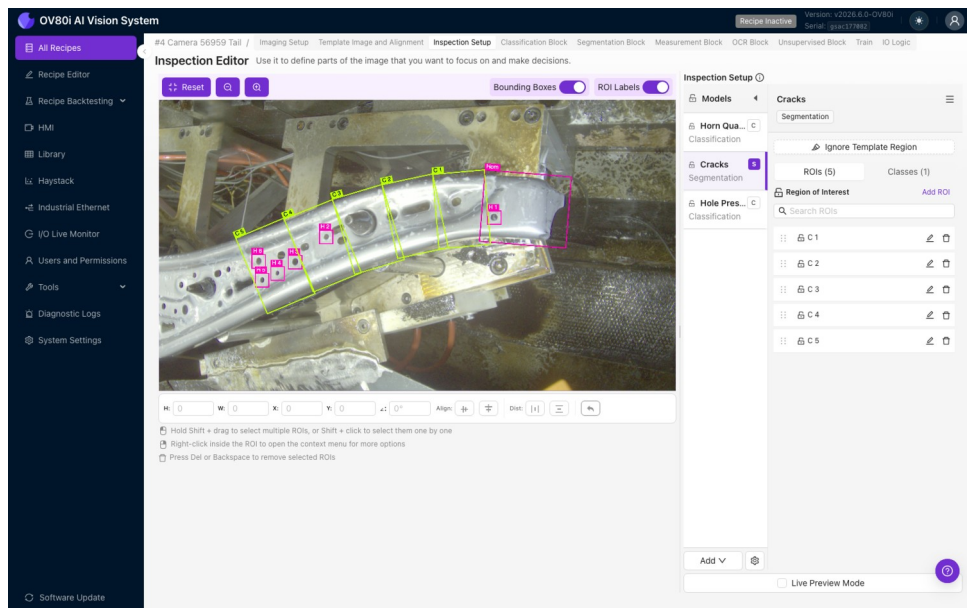
30

Training Images

—

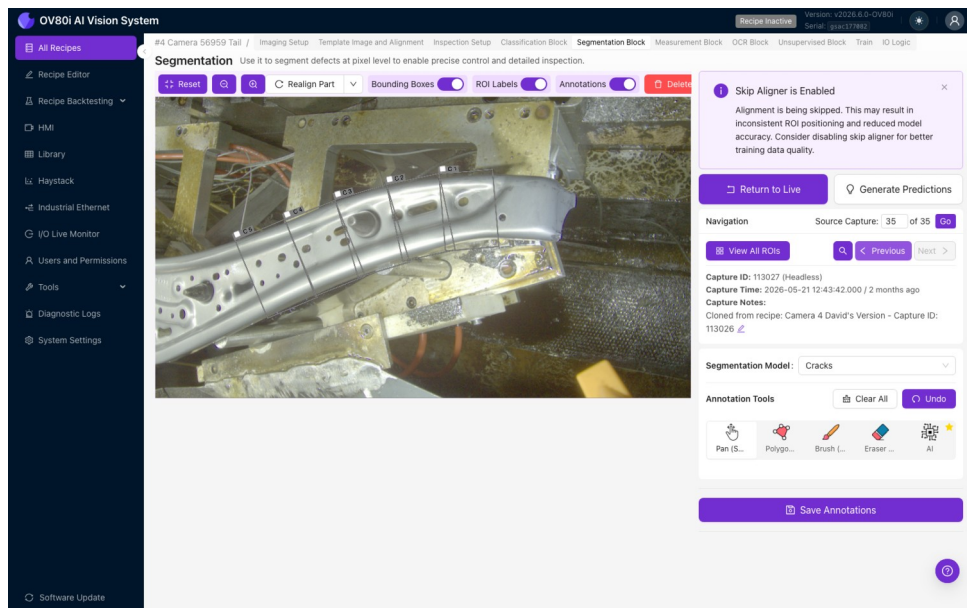
Deployment Time

Step 5: Setup Inspection Regions



Five regions, C 1 through C 5, tile the rail end to end. Together they cover the full length of the part for crack detection.

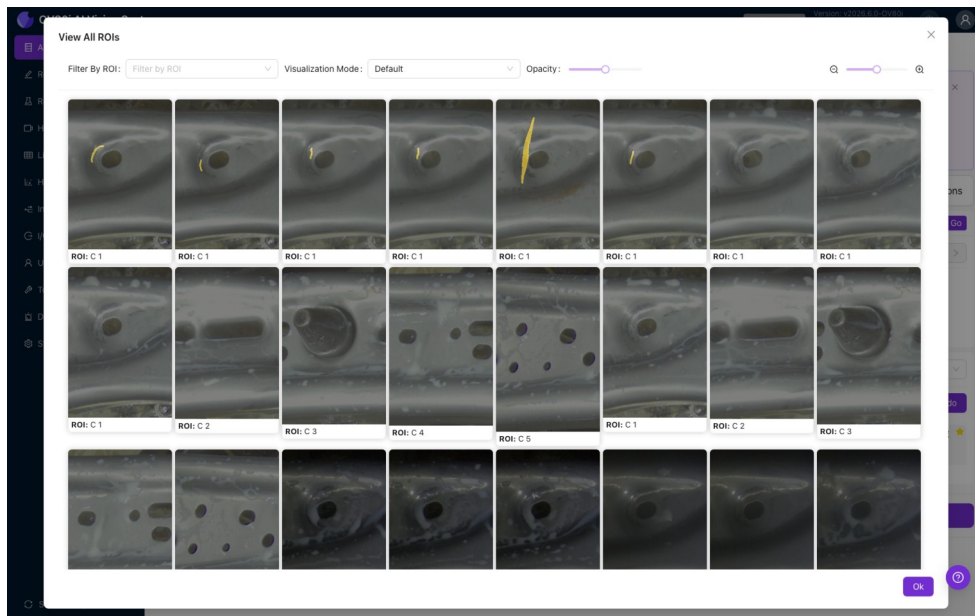
Step 6: Setup Segmenter inspection



A single-class segmentation model that marks cracks at pixel level. It paints the exact shape and size of each flaw so small, elongated defects near hole edges can be measured.

All labeled regions — Cracks

The gallery shows the collected crops from the five crack regions, roughly 24 visible at a time across C 1 to C 5. Several crops carry yellow brush annotations marking thin, elongated crack-like flaws near hole edges, while the rest are clean, unannotated good examples. These pixel annotations are what the crack model learns from.



Training settings — Cracks

Training mode: Accurate

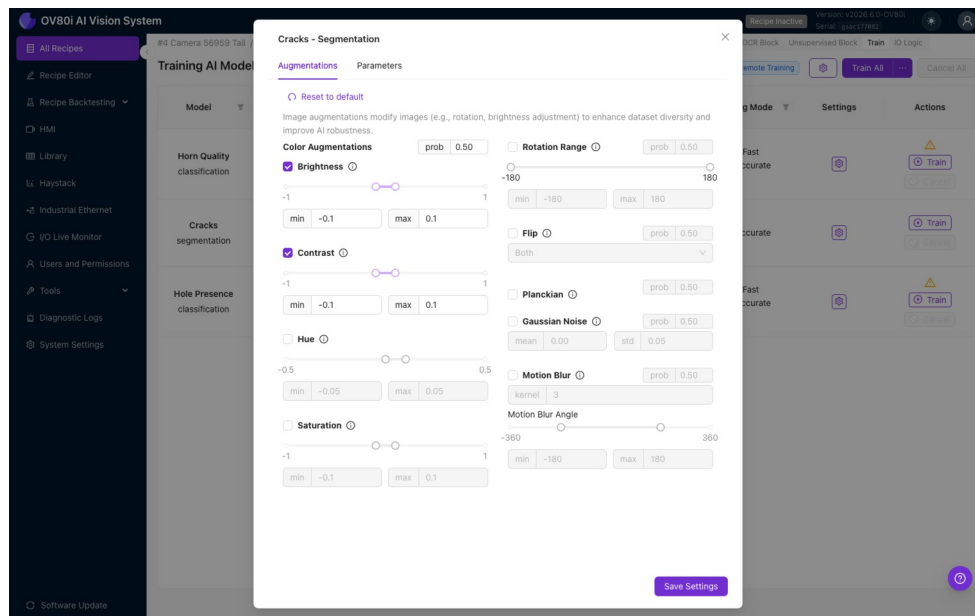
Color augmentation probability: 0.50

Brightness augmentation: enabled, -0.1 to 0.1

Contrast augmentation: enabled, -0.1 to 0.1

Hue and saturation: disabled

Rotation, flip, noise and motion blur: disabled



 OVERVIEW

Results

0.028
loss

Mean IoU

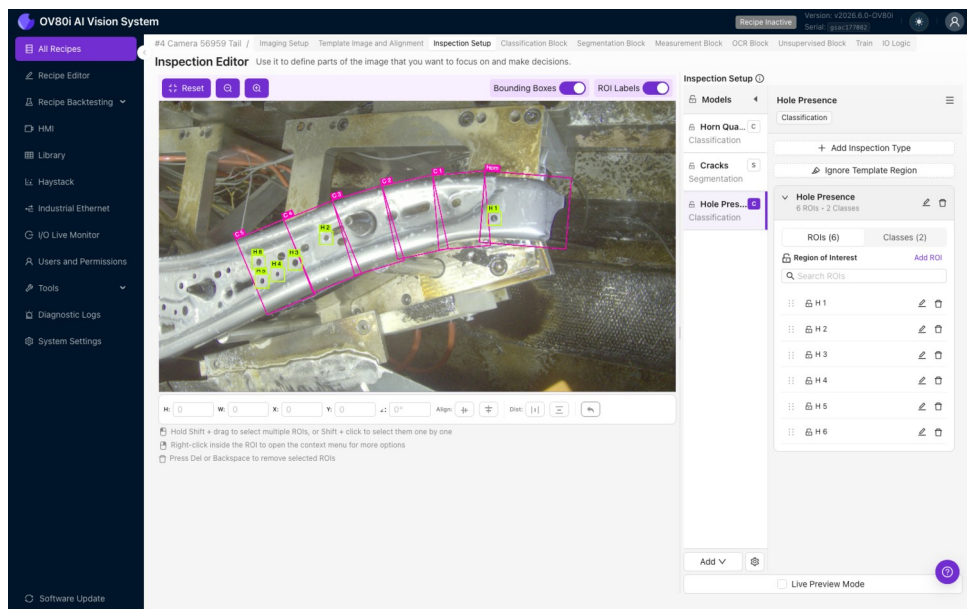
—

Training Images

—

Deployment Time

Step 7: Setup Inspection Regions



Six small regions, H 1 through H 6, sit one on each punched hole along the flange. Each is judged independently for hole presence.

Step 8: Setup classifier inspection

OV80i AI Vision System

Recipe inactive Version: v2028.6.5-OV80i Serial: 390177660

#4 Camera 56959 Tail / Imaging Setup Template Image and Alignment Inspection Setup **Classification Block** Segmentation Block Measurement Block OCR Block Unsupervised Block Train IO Logic

Classification
Use it to categorize an image into different classes based on its visual characteristics.

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Return to Live Generate Predictions

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Capture ID: 113028 (Headless)
Capture Time: 2026-05-21 12:16:37:000 / 2 months ago
Capture Notes:
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Inspection Types

Hole Presence - Hole Presence

Pass	pass_hole_presence	74
Fail	fail_hole_presence	9

Missing End - Horn Quality

Fail	fail_missing_end	7
Pass	pass_missing_end	23

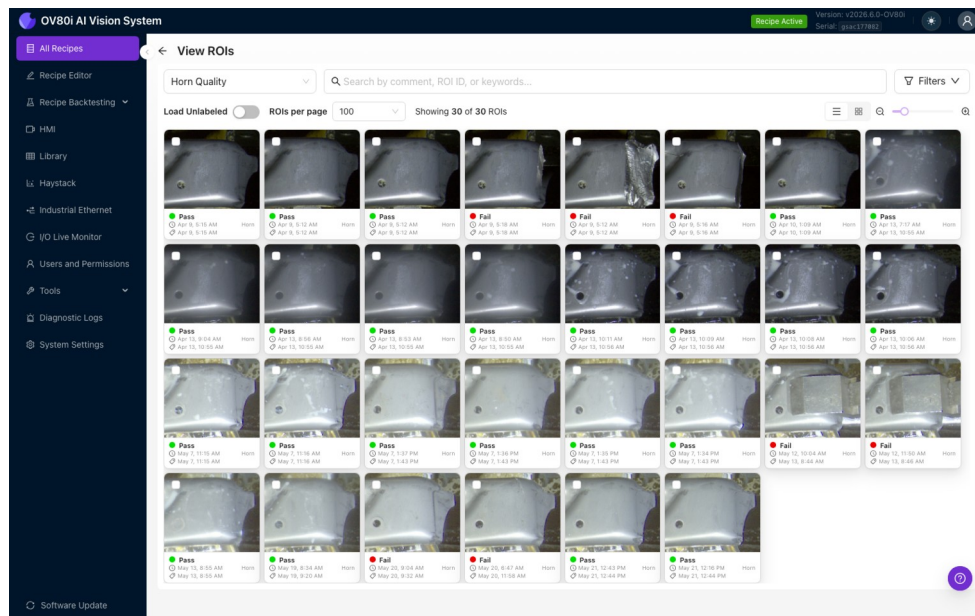
Train

Live Preview Mode Test

A two-class classifier deciding whether each hole is present. Classes are Pass (pass_hole_presence) and Fail (fail_hole_presence).

All labeled regions — Hole Presence

The gallery shows the labeled crops from the six hole regions, each tagged Pass or Fail. In total 74 crops are labeled `pass_hole_presence` and 9 `fail_hole_presence`, covering open holes as well as blank, unpunched surfaces. These labels define what counts as a present versus missing hole.



Training settings — Hole Presence

Training mode: Accurate

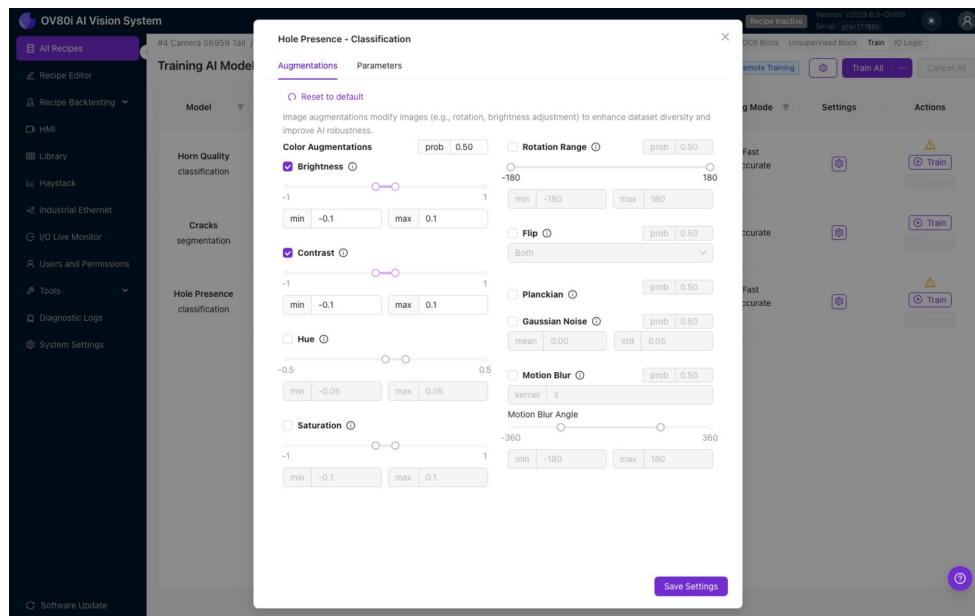
Color augmentation probability: 0.50

Brightness augmentation: enabled, -0.1 to 0.1

Contrast augmentation: enabled, -0.1 to 0.1

Hue and saturation: disabled

Rotation, flip, noise and motion blur: disabled



Training report — Hole Presence

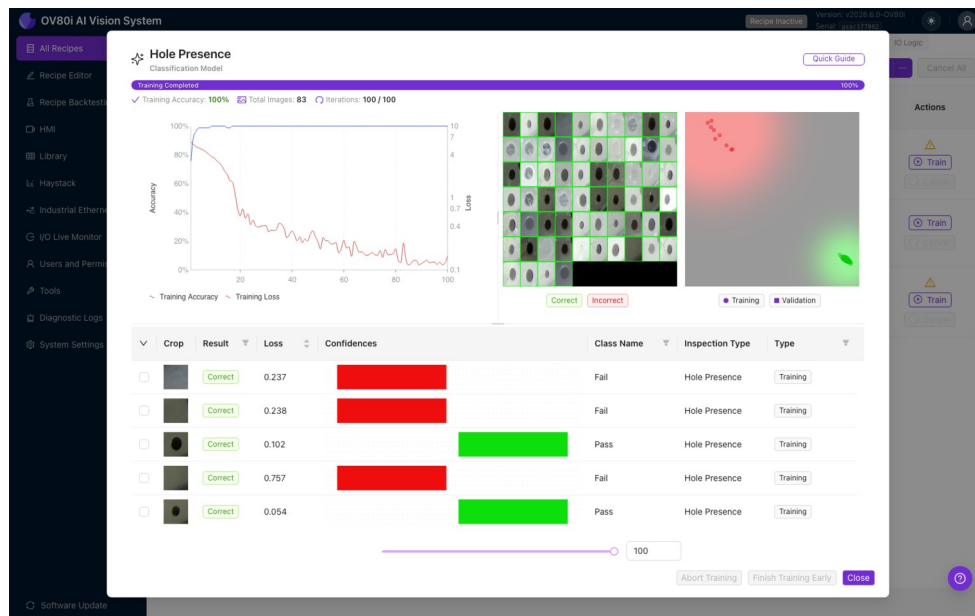
Training accuracy: 100%

Training loss: 0.154

Training images: 83

Iterations: 100 / 100

Labels: 74 pass_hole_presence, 9 fail_hole_presence



Results

100%

Training Accuracy

83

Training Images

—

Deployment Time

Step 9 Configure Node-Red

A part passes only if every classified region comes back Pass and no detected crack exceeds 250 pixels in area; any single failure fails the whole part. The verdict is written to the plant PLC over EtherNet/IP as a simple pass or fail signal on each cycle. A full result report, including the reason for every failure, is also sent to the cell's master controller for line-level tracking.

Library: Store >20,000 images on device.

Easier root cause analysis, faster retraining and redeploying. Easy production data retraining.

Library

Capture # 12345 Trigger ID 12345 Captured Between: 5m 30m 1h 12h 1d

Filter by Notes Filter by Recipe Filter by Pass / ...

> Advanced Filters: Metadata, Filter by Inspection Target, Inspection Region and Search Area

Include Trainset Images ☒

Alignment Fails Only ☐

Q Search 23051 Total Captures Sort By: Newest First

< 1 2 3 ... 1153 > 20 / page Go to Page Select All Delete All

#165488 70% real ng data + 4 synthetic (imported) 2026-07-27 19:49:44.493 a day ago **FAIL**

#165487 70% real ng data + 4 synthetic (imported) 2026-07-27 19:49:41.943 a day ago **FAIL**

#165486 70% real ng data + 4 synthetic (imported) 2026-07-27 19:49:39.369 a day ago **FAIL**

#165485 70% real ng data + 4 synthetic (imported) 2026-07-27 19:49:36.882 a day ago **FAIL**

#165484 70% real ng data + 4 synthetic (imported) 2026-07-27 19:49:34.263 a day ago **PASS**

#165483 70% real ng data + 4 synthetic (imported) 2026-07-27 19:49:31.102 a day ago **FAIL**

Capture #165488 from 70% real ng data + 4 synthetic (imported)

Trigger ID Q: 7000 Trigger Mode Q: Manual HMI Trigger Mode

Capture Time: 2026-07-27 19:49:44.493 a day ago

Process Time: 2421 ms

Download / Download (with labels) / Copy Link

Capture Notes: No notes for this capture.

Class Stats: ROI 36 - Inspection Type 1 - Search Area 1
Confidence Score Threshold: 0.5

Model	Class	# Blobs	Area (Pixels)
Model	Dent	0	0
Model	Discolor	1	2236
Model	Scratch	0	0

- Storage of training AND production data on the camera itself
- Able to leverage directly to retrain models

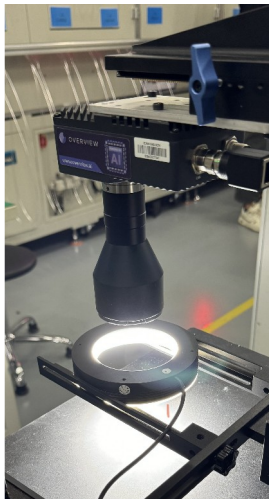


OVERVIEW

**Results: 1-2 hours to configure,
setup, label, train and deploy
each inspection model.**

The OV80i

Designed for 10x Faster Installs. The truly all-in-one Deep Learning AI Vision System.



Imaging
8.3MP Color



On-board Library and
Root-Cause Analysis
Tool (Haystack)



Powerful Computing
- Onboard Nvidia GPU



No software download
needed

Industry Leading Accuracy. Built on reliable, proven hardware. No software downloads or special computer requirements. *Easy setup*, fast ramp-up for engineers and technicians. Rapid troubleshooting and maintenance.

No software licenses, no recurring fees.

OV80i Benefits and Advantages

1. Industry leading AI accuracy. Form-factor and ease-of-use of vision sensor.
2. Reduced variance in manual human inspection, AI inspection is 100% inspection, 24/7.
3. Reduce escapes of bad parts.
4. Onboard data. Simplify root cause analysis and eventual yield gains.
5. Easy to train – AI models can be trained in 30 minutes or less by manufacturing, process, controls or quality engineers.
6. Flexibility – OV80i can be used for small component defect detection or larger assembly verification tasks.
7. Fast inspections – as low as 0.3 seconds per image.
8. Visual AI Inspections that can help replace human inspectors.
9. Multi-language Support.

5 factors make us unique



NVIDIA-Powered On-device training

Run **large AI models** and **retrain on the camera in minutes** to get **unmatched accuracy and performance on subtle defects**



Data-first with massive onboard storage

Labeling and training using **JPEG** for faster results and **storage for weeks worth of production data** (100k+ images)



Modern, browser-native tech stack

Access every camera instantly through a **simple browser** – **no installs, no license keys, and no dedicated PC** required



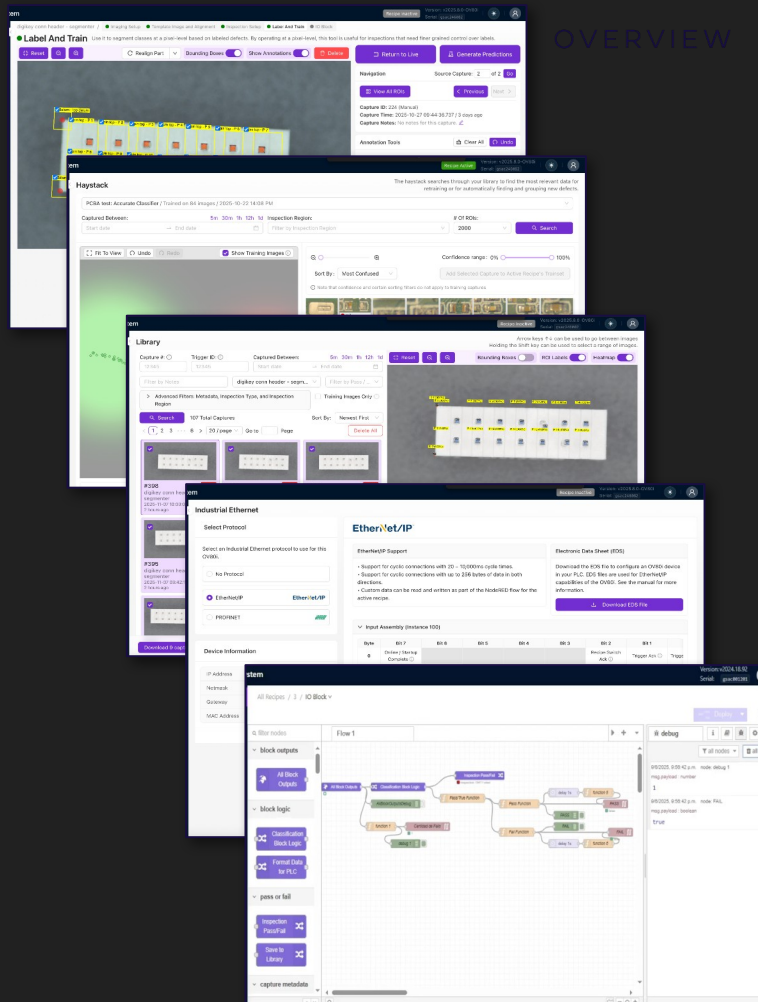
Root-cause, discovery defects, & continually improve

Use our proprietary **haystack** tool to easily **surface outliers & new defect modes**, and **retrain models using real production data**



Fully open automation platform (Node-Red + APIs)

Trigger, control, create dashboards, and **customize everything with APIs & Node-RED**, enabling **PLC-level logic** directly on the camera



OVERVIEW

Can't wait for the rare defect? Generate it.

Synthetic defects, guided by your real ones. A 12.4× faster path from new SKU to production-ready inspection.

12.4×

faster path from new SKU to production-ready inspection

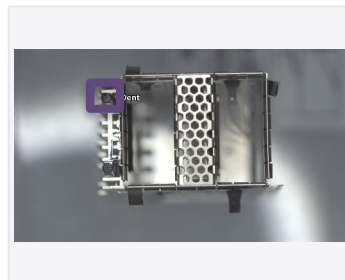
01

Capture good parts



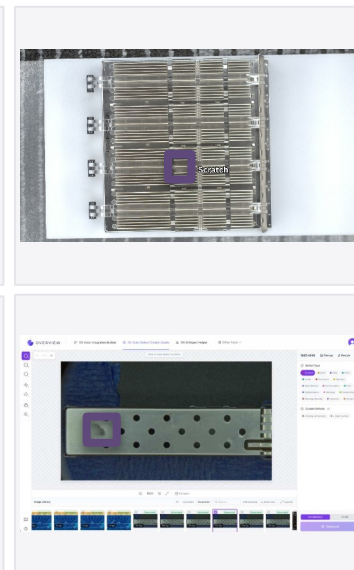
02

Generate defects



03

Train & verify



Integration built for everyone.

Controls engineers are some of the busiest people in the factory. Customize what works best for them, no code required.

Drag-and-drop logic

Anyone can author inspection logic without writing code.

PLC, MES, HMI

Bring real factory data into the camera and push results back out.

Custom dashboards

Generate the views and reports each engineer actually needs.

The screenshot displays the Overview OV Auto-Integration Builder interface. The top navigation bar includes the Overview logo and links to 'OV Auto-Integration Builder', 'OV Auto-Defect Creator Studio', 'OV AI Expert Helper', and 'Other Tools'. The main workspace is titled 'Test Data Editor' and features a 'Load saved data...' dropdown, 'Import: All Block Outputs', 'Save', and 'Trigger Simulated Capture' buttons. A sidebar on the left lists various data sources under categories like 'ocr', 'predictions', 'rol', 'rois', 'alignment', 'predictions', 'search_areas', 'measurement', 'segmentation', and 'classification'. The central workspace shows a 'Pin Tip Inspection' flowchart with blocks for 'AI Block Outputs', 'Get Serial from PLC', 'Inspect Pins & Contaminants', 'Inspection Pass/Fail', 'Save to Library', 'Capture Metadata', 'Status Banner', 'Pin Tip Map', and 'Pin Tip Stats Table'. The bottom status bar displays 'Inspection Result: FAIL', 'Digital IO Status: Output 1 NO RESULT, Output 2 NO RESULT', 'Save to Library: CAPTURE SAVED', 'Onboard Status LED: No Result', 'Capture Metadata: pin_serial_number: "EN-000740837", result: "FAIL", tip_count: 80', and a 'Dashboard' section with a 'FAIL' status and a 'Click to expand' button.

Remote personnel — Worldwide

United States

San Francisco, CA
Columbus, OH
Nashville, TN
Seattle, WA
Iowa City, IA
Atlanta, GA
Charlotte, NC
Detroit, MI
Austin, TX

Americas

CDMX, MX
Guadalajara, MX
Hermosillo, MX
Curitiba, Brazil

Asia

Taipei, TW
Shanghai, CH
Nantong, CH
Shenzhen, CH
Haiyan, CH
Ha Noi, VN
Bengaluru, IN

Europe

Zurich, Switzerland



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OVERVIEW

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